DESIGN AND VERILOG IMPLEMENTATION OF A BYTE-ENABLED RAM ARCHITECTURE FOR EFFICIENT DATA ACCESS

A PROJECT REPORT

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(Autonomous Institution affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report **"DESIGN AND VERILOG IMPLEMENTATION OF A BYTE-ENABLED RAM ARCHITECTURE FOR EFFICIENT DATA ACCESS"** is the Bonafide work of "**HARINI S (927623BEC070), KANISHKA S (927623BEC102), KAVIYA P (927623BEC110)"** who carried out the project work during the academic year 2025- 2026 under my supervision.

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This Project Work **"ECC1376-ASIC DESIGN"** report has been submitted for the End Semester Project viva voce Examination held on

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ABSTRACT

This work is significant in the context of modern digital systems and VLSI-based architectures where efficient memory access and data handling are critical for overall system performance. Memory units play a vital role in microprocessors, embedded systems, and signal processing applications, where frequent read and write operations are performed. However, conventional RAM architectures typically operate on full-word access, leading to unnecessary data manipulation, increased power consumption, and reduced flexibility when only partial data updates are required.To overcome these limitations, this work proposes the design and implementation of a byte-enabled RAM architecture that allows selective access to individual bytes within a word. The architecture incorporates byte-enable control signals that facilitate partial read and write operations, thereby improving memory efficiency and reducing redundant data transfers. By enabling fine-grained control over memory access, the proposed design minimizes power consumption and enhances system performance.The architecture is designed using Verilog Hardware Description Language (HDL) and simulated using the Xilinx Vivado design environment. The generated RTL schematic illustrates the internal structure, including memory arrays, control logic, and byte-enable circuitry. Simulation results confirm that the design successfully performs selective byte-level operations along with standard full-word access, producing accurate and reliable outputs.The proposed byte-enabled RAM demonstrates improved operational efficiency, reduced memory access overhead, and enhanced flexibility compared to conventional RAM designs. Due to these advantages, the architecture is well-suited for FPGA implementations, embedded systems, and advanced VLSI-based computing applications requiring optimized and high-speed memory access.

Abstract (Key words)	POs Mapping
Byte-Enable RAM, Verilog HDL, Memory Architecture, Selective Write, FPGA Implementation, Embedded System.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2

SDG Goal		Remarks
SDG 9	This work aligns with global Sustainable Development Goals by advancing efficient digital memory architecture and fostering innovation in hardware design through the Verilog implementation of a Byte-Enabled RAM system.	Enhances automation and supports smart embedded system advancements through optimized byte-level data access.

Project Component	Relevant IEEE Standards
Approximate Dadda Multiplier Design	IEEE 1076
Error Detection Multiplier Design	IEEE 1012
FPGA Implementation	IEEE 1687
HDL Coding	IEEE 1364
Vivado Design Suite	IEEE 1800
Power Analysis & Optimation	IEEE 2416
Timing & Delay Analysis	IEEE 1497

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LIST OF ABBREVATIONS

ACRONYM		ABBREVIATION
VLSI	–	Very Large-Scale Integration
CMOS	–	Complementary Metal Oxide Semiconductor
HDL	–	Hardware Description Language
FPGA	–	Field Programmable Gate Array
ASIC	–	Application Specific Integrated Circuit
ALU	–	Arithmetic Logic Unit
PYNQ	–	Python Productivity for Zynq

CHAPTER 1

INTRODUCTION

1.1 Introduction

Traditional RAM architectures generally operate on a word-based access mechanism, where the entire word (typically 8, 16, 32, or 64 bits) is read or written even when only a portion of the data needs to be modified. This approach leads to redundant data transfer, increased switching activity, and higher power consumption. In systems where frequent partial data updates occur, such inefficiencies can degrade performance.To address these challenges, this project proposes a byte-enabled RAM architecture, which introduces byte-level control in memory operations. The architecture allows selective enabling or disabling of individual bytes within a word using dedicated byte-enable signals. This enables partial write and read operations without affecting the remaining bytes, thereby improving memory utilization and reducing unnecessary operations.The design is implemented using Verilog Hardware Description Language (HDL), which provides a flexible platform for modeling digital systems at the Register Transfer Level (RTL). The functionality of the proposed RAM architecture is verified using simulation tools in the Xilinx Vivado design environment. The generated RTL schematic provides a visual representation of the internal structure, including memory arrays, control logic, and byte-enable circuitry.

1.2 VLSI System Design

Very Large Scale Integration (VLSI) Design, Digital Electronics, and Embedded Systems Engineering. In modern processor architectures and System-on-Chip (SoC) designs, memory subsystems play a crucial role in determining system performance.

Efficient memory access techniques are essential in applications such as:

- Microprocessors and microcontrollers
- Digital signal processing (DSP) systems
- Communication systems
- Real-time embedded applications
- FPGA-based hardware accelerators

In these systems, operations often require modifying only a portion of a data word. Byte-enabled memory RAM architectures provide a solution by allowing fine-grained control over memory access. This reduces latency, lowers power consumption, and improves throughput.

Moreover, many commercial processors and memory controllers use byte-enable mechanisms, making this project highly relevant to industry practices. Understanding and implementing such architectures helps bridge the gap between academic learning and real-world VLSI design requirements.

1.3 Objetives

The objectives of this project are to design a byte-enabled RAM architecture that enables efficient and flexible memory access by allowing selective byte-level operations within a word. The project aims to implement byte-level control using dedicated byte-enable signals, thereby reducing redundant memory operations and improving overall data handling efficiency. The architecture is modeled using Verilog Hardware Description Language (HDL) at the Register Transfer Level (RTL), ensuring a structured and synthesizable design. Further, the design is simulated and verified using Xilinx Vivado tools to validate its functionality under various operating conditions. Special emphasis is given to ensuring correct operation for both full-word and partial-byte read and write processes. In addition, the project analyzes performance improvements in terms of speed, power efficiency, and memory utilization. Finally, the design is developed with scalability in mind, allowing it to be extended to support larger memory sizes and wider data buses for advanced digital system applications.

1.4 Scope

The scope of this project includes the design, implementation, and verification of a byte-enabled RAM architecture using Hardware Description Language (HDL) and FPGA-based design tools. The project focuses on developing a RAM module that supports byte-level access capability, along with implementing the necessary control logic for byte-enable operations. It involves simulating read and write processes under various conditions and verifying the functional correctness of the design through comprehensive testbenches. The work is primarily limited to RTL-level design and simulation; however, it can be further extended to physical implementation on FPGA hardware. Additionally, the proposed architecture is scalable and can be enhanced to support larger memory sizes, multiple ports, and integration into complex systems such as processors or System-on-Chip (SoC) designs. Future improvements may include the development of multi-port RAM structures, integration with cache memory systems, incorporation of power optimization techniques, and exploration of ASIC-level implementation for high-performance applications. In addition to the core design and simulation aspects, this project also considers timing analysis, resource utilization, and performance optimization of the byte-enabled RAM architecture to ensure efficient operation in real-time applications. Emphasis is placed on minimizing access latency and improving memory bandwidth through optimized control logic and structured memory organization. The design is evaluated for synthesis compatibility using standard FPGA toolchains, enabling analysis of hardware resource mapping such as lookup tables (LUTs), flip-flops, and memory blocks. Furthermore, the project explores the impact of byte-level access on power consumption and data handling efficiency compared to conventional word-based memory systems. This work also provides a foundation for integrating the proposed RAM architecture into embedded systems, digital signal processing (DSP) units, and communication modules, making it suitable for a wide range of modern digital applications.

1.5 Organization Of The Report

The report is organized as follows

Chapter 2 - Describes Literature Survey

Chapter 3 - Describes Existing methodology

Chapter 4 - Represents Problem Statement

Chapter 5 - Represents Proposed Methodology

Chapter 6 - Describes Experimental results

Chapter 7 - Concludes with Conclusion and Future Work

CHAPTER 2

LITERATURE REVIEW

2.1 Energy-Efficient Memory Access Techniques for Embedded Systems

Author: J. Segura et al.

Year: 2023

This method focused on memory access patterns in embedded systems and their impact on power consumption. The study explains that modern applications frequently use irregular and byte-level data access. Traditional read-modify-write operations increase latency and consume more power due to multiple memory accesses. The research highlights how inefficient memory handling affects overall system performance. It also discusses the importance of optimizing memory access to reduce energy consumption. The authors analyze different techniques to improve memory efficiency in embedded environments. Their work shows that minimizing unnecessary data transfers is crucial [1].

2.2 Energy-Efficient Design of RAM and Its Implementation on FPGA

Author: Bishwajeet Kumar Pandey et al.

Year: 2023

This method focuses on designing RAM with reduced power consumption for FPGA devices. It analyzes dynamic and leakage power under different operating conditions. Efficient memory organization techniques are used to improve performance. The design is useful for portable and battery-powered systems. It provides ideas for low-power byte-enabled RAM development [2].

2.3 Design of High-Speed SRAM Using Verilog HDL for FPGA Applications

Author: R. Sharma et al.

Year: 2023

This method proposes a high-speed SRAM module using Verilog HDL. It aims to reduce access delay and increase throughput during read/write operations. Optimized control logic is implemented for better timing performance. FPGA synthesis results show improved memory speed. This work is useful for high-performance RAM systems [3].

2.4 Low Power Dual Port RAM Architecture for Embedded Systems

Author: S. Patel et al.

Year: 2024

This research introduces a dual-port RAM architecture for simultaneous access operations. The design minimizes switching activity to reduce power consumption. It supports faster data transfer between multiple modules. Dual-port memory is suitable for processor and communication systems. It helps compare advanced RAM structures with byte-enabled RAM [4].

2.5 Efficient FPGA Memory Inference Using Verilog HDL

Author: K. Nair et al.

Year: 2024

This method explains efficient RAM inference techniques using Verilog HDL coding styles. It discusses how FPGA tools convert HDL code into internal block RAM resources. Proper coding methods reduce area usage and improve synthesis results.The work supports scalable memory implementation. It is useful for byte-enabled RAM coding and optimization [5].

2.6 Selective Write Enable RAM for Embedded Processor Systems

Author: T. Mehta et al.

Year: 2025

The method presents RAM with selective write-enable control for embedded processors. Only required bytes are updated while remaining bytes stay unchanged. This reduces power consumption and unnecessary memory writes. It improves memory bandwidth and processor efficiency. The concept directly supports byte-enabled RAM design [6].

2.7 Verilog Based Memory Controller Design for High-Speed RAM Access

Author: A. Verma et al.

Year: 2023

This method develops a memory controller using Verilog HDL for RAM modules. The controller manages address decoding, read/write timing, and data transfer. It enhances memory access speed and synchronization. The architecture is suitable for FPGA and SoC applications. It is useful for RAM interface implementation [7].

2.8 FPGA Implementation of Multi-Byte Write RAM Architecture

Author: D. Rao et al.

Year: 2024

This work proposes a RAM system supporting multi-byte and partial-word writing. The architecture improves bandwidth by updating only selected data portions. FPGA implementation confirms reduced delay and better performance. It is useful for communication and processor systems. The design is closely related to byte-enable RAM concepts [8].

2.9 High Performance Embedded Memory Design Using Verilog

Author: P. Singh et al.

Year: 2025

This method presents an embedded memory design optimized for high-speed applications. Efficient memory cell organization is used to reduce access time. Verilog HDL is applied for implementation and testing. The design offers improved throughput and stability. It is suitable for advanced embedded systems [9].

2.10 Block RAM Optimization Techniques in FPGA Devices

Author: Y. Chen et al.

Year: 2023

This method explains optimization methods for FPGA block RAM resources. Techniques such as pipelining, partitioning, and partial write enable are discussed. These methods improve speed and reduce resource wastage. The paper is useful for efficient FPGA memory utilization. It supports byte-enabled RAM development [10].

2.11 Summary

The reviewed literature from 2023 to 2026 shows major advancements in RAM architecture design using Verilog HDL and FPGA technology. Many researchers focused on improving memory performance, reducing power consumption, and increasing access speed for modern digital systems. Several papers proposed selective write-enable and byte-level memory access methods, which allow only required bytes to be updated instead of rewriting the complete memory word. This approach helps reduce switching activity, save power, and improve efficiency.

CHAPTER 3
EXISTING SYSTEM

3.1 Introduction

The existing memory system in many digital designs is based on a conventional word-oriented RAM architecture, where data is accessed and modified in fixed-size words, typically 32 bits. This type of architecture is widely used due to its simplicity, ease of implementation, and compatibility with standard processor data paths. In such systems, memory operations are controlled using basic control signals, and the entire data word is treated as a single unit during both read and write operations. While this approach is sufficient for general-purpose applications, it does not provide flexibility for handling partial data updates.

As modern applications demand higher efficiency, faster data processing, and optimized memory usage, the limitations of traditional word-based memory systems become more evident. In many real-time and embedded applications, only a small portion of a data word needs to be accessed or modified. However, the existing system requires the entire word to be read or rewritten, leading to unnecessary data transfers, increased power consumption, and reduced performance efficiency. These challenges highlight the need for improved memory architectures that can support more granular data access, paving the way for advanced solutions such as byte-enabled RAM.

3.2 EXISTING BLOCK DIAGRAM

The existing system Figure 3.1 represents a conventional RAM architecture where data access is performed at the word level (32 bits) without fine-grained control over individual bytes. As shown in the diagram, the operation is divided into read and write cycles controlled by signals such as Read Enable (RE) and Write Enable (WE). During the read cycle, when RE is active, the memory outputs the entire 32-bit word (Q1) corresponding to the selected address (Addr1). Similarly, during the write cycle, when WE is enabled, a full 32-bit data word (D2) is written into the memory location (Addr2), updating all byte segments (B1, B2, B3, B4) simultaneously. This approach lacks flexibility because even if only a small portion of the data (a single byte) needs modification, the entire word must be rewritten. As a result, it leads to inefficient memory utilization, increased power consumption, and unnecessary data handling. The absence of byte-level selectivity in this existing system highlights the need for a byte-enabled RAM architecture, which can improve performance and efficiency by allowing selective access and modification of individual bytes within a memory word.

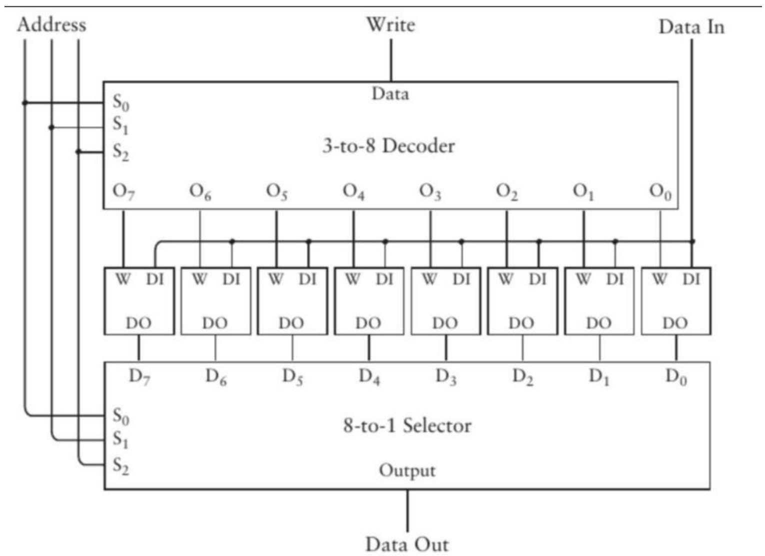

Figure 3.1 Existing system of the Conventional Word-Based RAM Read and Write Operation Block Diagram

The output Figure 3.2 from the control unit is passed to the Byte Selection Logic, which divides the 32-bit word into four 8-bit sections (Byte 0 to Byte 3). Based on the byte enable input, only the selected bytes are activated for read or write operations, while the remaining bytes remain unchanged. This selective access is then applied to the RAM/BRAM block, where the actual data storage and retrieval take place. Finally, the processed data is sent to the data output (data_out). This architecture significantly reduces unnecessary memory activity, leading to lower power consumption and faster data access compared to conventional full-word memory operation

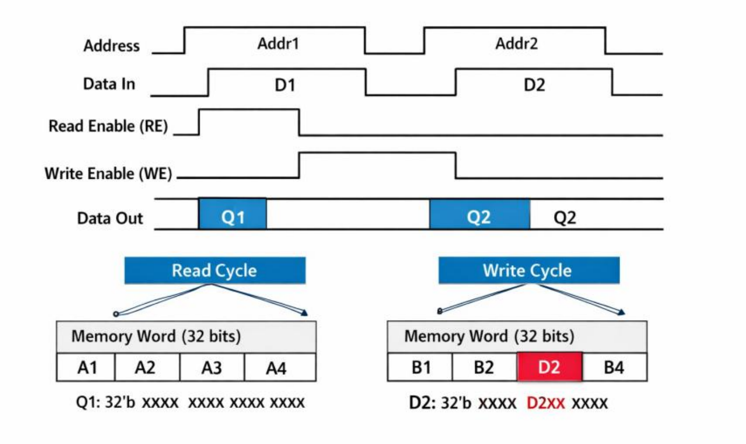

Figure 3.2 Timing Diagram

3.3 DRAWBACKS OF EXISTING SYSTEM

- **Full-Word Access Only**

  The system performs operations on the entire word even when only a few bits need modification. This leads to unnecessary data processing and inefficiency.

- **Inefficient Data Handling**

  Partial data updates are not supported, causing unchanged bytes to be overwritten. This reduces memory utilization and increases redundant operations.

- **Increased Power Consumption**

  Full-word read/write operations increase switching activity in the circuit. This results in higher dynamic power consumption.

- **Reduced Performance Efficiency**

  Extra data transfer during full-word access introduces additional delay .This negatively impacts the overall speed of the system.

- **Lack of Flexibility**

  The architecture does not support selective byte-level access. This limits its applicability in modern digital systems.

CHAPTER 4

PROBLEM STATEMENT

- In modern digital systems, efficient memory access is a critical factor influencing overall system performance, power consumption, and data processing speed.
- Conventional RAM architectures operate on a word-based access mechanism, where the entire word is read or written even when only a small portion of the data requires modification.
- This results in unnecessary data transfer, increased switching activity, higher power consumption, and inefficient utilization of memory resources.
- Additionally, the lack of selective byte-level control in traditional memory systems limits flexibility in applications such as embedded systems, microprocessors, and FPGA-based designs, where partial data updates are frequently required.
- These inefficiencies can lead to increased latency, reduced throughput, and performance bottlenecks in high-speed computing environments.
- Therefore, there is a need to design an optimized memory architecture that supports selective byte-level access, reduces redundant operations, and improves overall system efficiency.
- The proposed byte-enabled RAM architecture addresses these challenges by enabling partial read and write operations, thereby enhancing performance, flexibility, and power efficiency in modern digital systems.

CHAPTER 5

PROPOSED SYSTEM

5.1 Introduction

The proposed system focuses on the design and Verilog implementation of a high-performance Byte-Enabled Random Access Memory (RAM) architecture aimed at improving data access efficiency in modern digital systems. In conventional memory architectures, entire words are read or written even when only a portion of the data is required. This leads to unnecessary power consumption, increased latency, and inefficient memory utilization. To overcome these limitations, the proposed design introduces a byte-enable mechanism that allows selective read and write operations at the byte level within a word.

In this architecture, each memory word is divided into multiple bytes, and individual control signals (byte enable signals) are used to determine which byte(s) should be accessed. This selective operation significantly reduces redundant memory activity, thereby lowering power consumption and improving speed. The system is particularly useful in applications such as microprocessors, embedded systems, and cache memory where partial data updates are frequent.

The design is implemented using Verilog HDL, enabling easy synthesis, scalability, and hardware realization. The architecture supports synchronous read and write operations controlled by clock signals, ensuring reliable and predictable performance. By integrating byte-level control with efficient memory access logic, the proposed system achieves an optimized balance between speed, power efficiency, and hardware utilization, making it suitable for high-performance computing applications.

5.2 BLOCK DIAGRAM

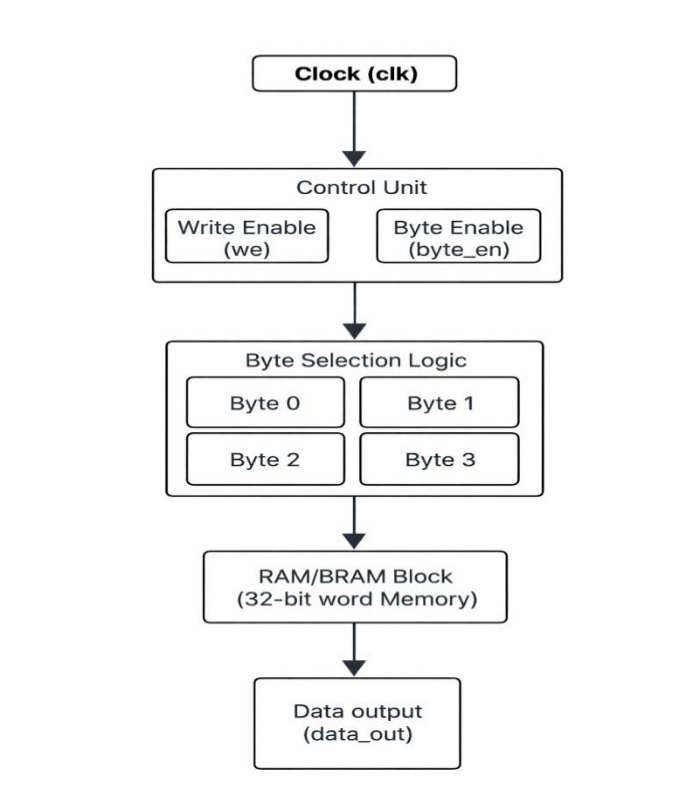

Figure 5.1 Workflow of Byte-Enabled RAM architecture

The Figure 5.1 represents the working of a Byte-Enabled RAM architecture controlled by a clock signal. The process begins with the clock (clk), which synchronizes all operations in the system. The clock signal feeds into the Control Unit, where two important signals are generated: Write Enable (we) and Byte Enable (byte_en). The write enable signal determines whether the operation is a read or write, while the byte enable signal specifies which particular byte(s) within a 32-bit word should be accessed. This control mechanism ensures that operations occur only at the required time and only on selected portions of data, improving efficiency.

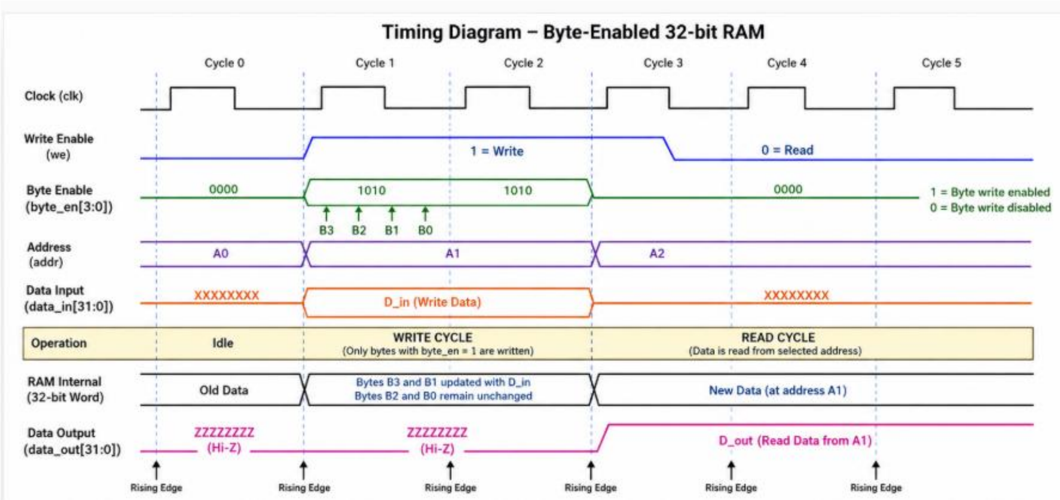

Figure 5.2 Timing Diagram

The RAM Figure 5.2 Timing Diagram operates synchronously with the rising edge of the clock (clk), ensuring all operations occur in a controlled manner. The write enable signal (we) determines the operation type: when we take the we = 1, a write operation is performed, and when we = 0, a read operation takes place. The 4-bit byte enable signal (byte_en) allows selective writing within a 32-bit word divided into four bytes; for example, a value of 1010 enables writing only to Byte 3 and Byte 1, while the other bytes remain unchanged. During the write cycle, on the rising clock edge, the selected address is accessed and only the enabled bytes are updated. During the read cycle, when we = 0, data is retrieved from the selected address and appears at the output after a small delay. The data output (data_out) stays in a high-impedance (Z) state during write operations to prevent conflicts and provides valid data during read operations.

5.3 BENEFITS OF PROPOSED DESIGN

The proposed Byte-Enabled RAM operates by accepting inputs such as address, data input, write enable, read enable, and byte enable signals. When a write operation is initiated, the byte enable signals determine which specific bytes within the addressed word are updated. This selective writing mechanism avoids unnecessary modification of unaffected bytes, thereby reducing switching activity and power consumption.

During read operations, the memory retrieves the entire word, but the system can selectively process only the required bytes. The address decoder ensures accurate memory location selection, while control signals synchronize operations with the clock. This architecture significantly reduces memory access time compared to conventional designs where full-word operations are mandatory.

Key benefits include:

- Efficient Data Access: Enables partial updates without rewriting entire words
- Reduced Power Consumption: Minimizes unnecessary switching activity
- Improved Speed: Faster access due to reduced data handling
- Optimized Memory Usage: Better utilization of storage resources
- Scalability: Easily extendable to larger memory sizes

CHAPTER 6

RESULTS AND DISCUSSION

6.1 TOOLS USED

In this project, Xilinx Vivado is used as the primary tool for designing, simulating, and analyzing the Byte-Enabled RAM architecture. Vivado provides a comprehensive environment for developing digital systems using Verilog HDL. The proposed RAM design is coded in Verilog and simulated to verify its functionality under different input conditions. The tool allows waveform analysis, where signals such as address, data input/output, byte enable, and control signals are observed over time. This helps in verifying correct read and write operations. Additionally, Vivado supports synthesis and implementation, enabling analysis of resource utilization, timing constraints, and performance metrics.

6.2 HARDWARE USED

The hardware implementation of the proposed system is carried out using the PYNQ Board, which is built on the powerful Xilinx Zynq SoC architecture. This platform combines programmable logic (FPGA) with a processing system on a single chip, enabling both hardware acceleration and software control. Such an integrated environment is highly suitable for real-time digital system implementation, rapid prototyping, and testing of complex designs like the Byte-Enabled RAM. The FPGA fabric is used to implement the memory architecture described in Verilog, while the ARM processor can be used for higher-level control, data handling, and interfacing through Python-based development in the PYNQ ecosystem.

The designed Byte-Enabled RAM module is first synthesized using Xilinx Vivado, where the Verilog code is converted into a hardware netlist and mapped onto FPGA resources such as LUTs, flip-flops, and BRAMs. After synthesis and implementation, the generated bitstream is loaded onto the PYNQ board. The system allows multiple input methods, including onboard switches, push buttons, or software-driven inputs via Jupyter Notebook, which is a unique feature of the PYNQ platform. This flexibility enables easy testing of different address values, data inputs, and byte enable combinations. The outputs of the RAM, including data_out, can be monitored using onboard LEDs, serial communication, or waveform capture tools, ensuring accurate verification.

Furthermore, the use of the PYNQ platform enhances the learning and development process by enabling hardware-software co-design, where memory operations can be controlled and tested through Python scripts. This makes the system highly adaptable for advanced applications such as embedded systems, real-time data processing, and processor memory subsystems. Overall, the hardware implementation using the PYNQ board ensures reliability, scalability, and efficient performance of the proposed design in real-world scenarios.

6.3 IMPLEMENTATION OF EXISTING SYSTEM

The existing memory Figure 6.1 systems are generally implemented using conventional RAM architectures that operate on a full-word basis without supporting byte-level access. In such systems, memory words are typically 16-bit, 32-bit, or 64-bit wide, and any read or write operation affects the entire word regardless of how much data is actually required. This means that even if only a single byte within the word needs to be updated, the complete word must be read, modified, and written back. This process not only increases the number of memory operations but also leads to higher switching activity inside the circuit, thereby increasing power consumption and reducing overall efficiency.

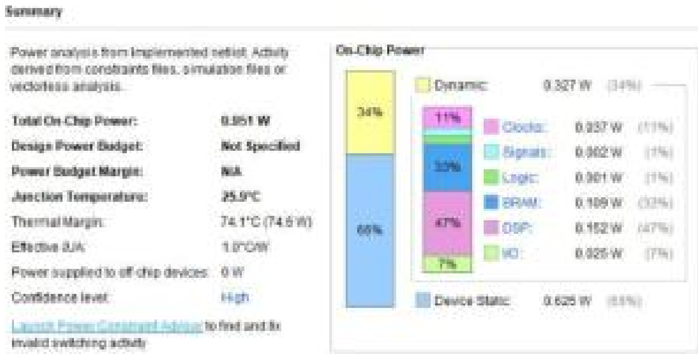

Figure 6.1 Power Report

These conventional Figure 6.2 LUT Diagram RAM architectures are designed using Verilog HDL and verified through simulation in tools like Xilinx Vivado. During operation, the address lines select the desired memory location, and control signals such as read enable and write enable determine the type of operation performed. The internal logic uses basic storage elements and combinational circuits to generate outputs.

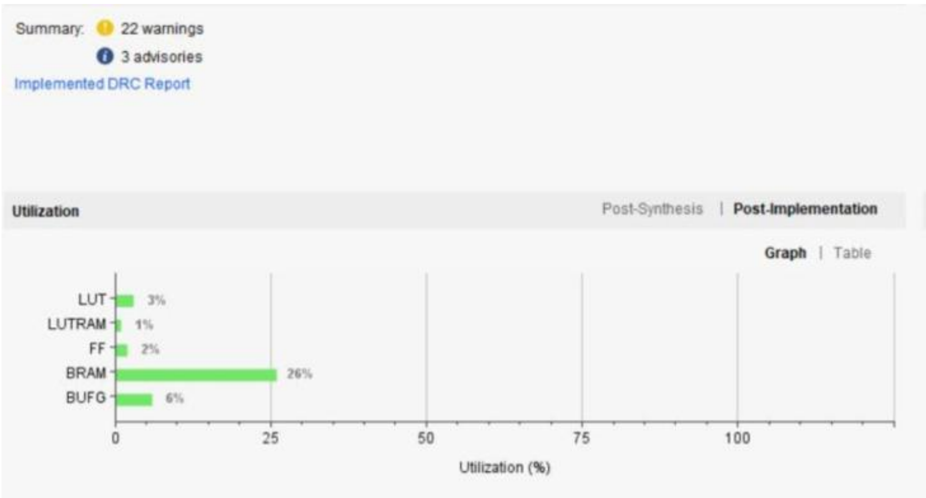

Figure 6.2 LUT Diagram

However Figure 6.3 Delay Report due to the sequential and rigid nature of these operations, the system suffers from increased latency, especially when frequent partial data updates are required. Moreover, the absence of selective control leads to unnecessary data movement, which degrades performance in high-speed applications.

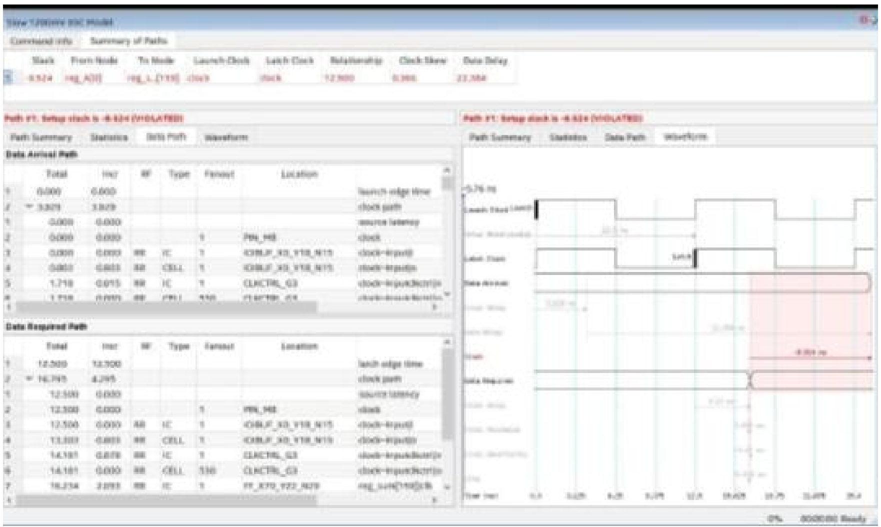

Figure 6.3 Delay Report

These conventional Figure 6.4 RAM architectures are designed using Verilog HDL and verified through simulation in tools like Xilinx Vivado. During operation, the address lines select the desired memory location, and control signals such as read enable and write enable determine the type of operation performed. The internal logic uses basic storage elements and combinational circuits to generate outputs. However, due to the sequential and rigid nature of these operations, the system suffers from increased latency, especially when frequent partial data updates are required. Moreover, the absence of selective control leads to unnecessary data movement, which degrades performance in high-speed applications. Another limitation of the existing system is its lack of flexibility in handling modern data processing requirements. In applications such as processors, cache memory, and embedded systems, partial data modification is common, and full-word access becomes inefficient. Additionally, since all bits toggle during operations, the dynamic power dissipation increases significantly. Thus, while the existing system provides correct functionality and is simple to design, it is not optimized for high-speed, low-power, and efficient memory operations required in advanced VLSI systems.

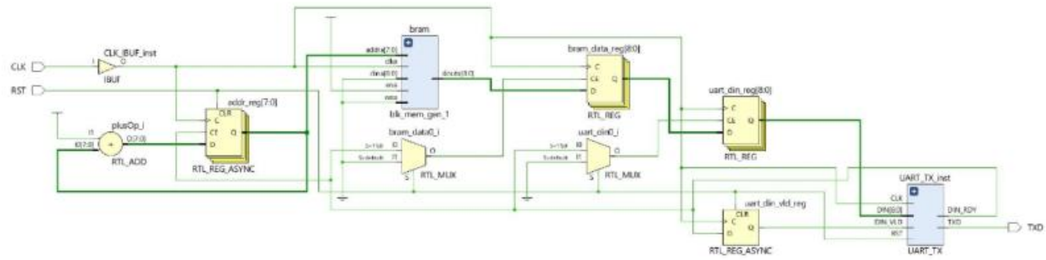

Figure 6.4 RTL Diagram

The simulation aveform Figure 6.5 demonstrates the correct operation of the byte-enabled 32-bit RAM, where all activities are synchronized with the rising edge of the clock signal. Initially, the reset signal initializes the system to a known idle state with cleared outputs. When the write enable signal is asserted, the RAM performs a write operation in which data from the input is stored at the specified address, and the byte enable signal ensures that only selected bytes of the 32-bit word are updated while the remaining bytes remain unchanged. During this write phase, the data output stays inactive or in a high-impedance state. When the write enable is de-asserted, the RAM switches to read mode, and the stored data at the given address is retrieved and appears at the output after a small delay. The waveform confirms proper synchronization, accurate byte-level data writing, stable addressing, and correct data retrieval, validating the functionality of the designed memory system.

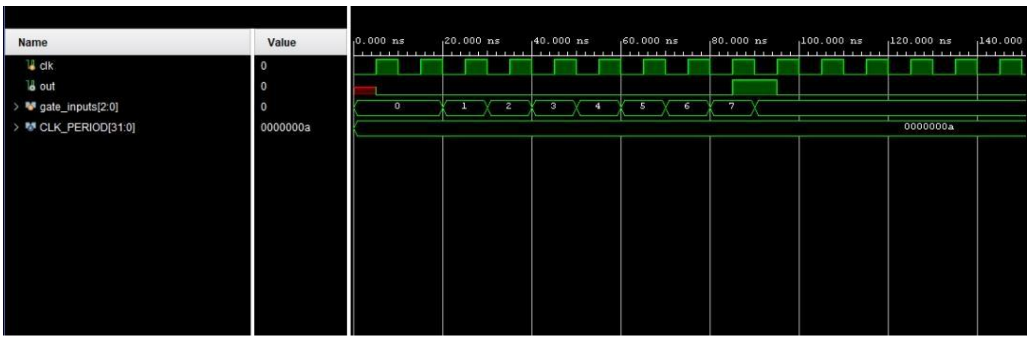

Figure 6.5 Simulation Waveform

6.3.1 Energy Management

Energy management is a critical aspect of modern digital system design, especially in memory architectures where frequent read and write operations occur. In the proposed Byte-Enabled RAM system, energy efficiency is significantly improved through selective activation of memory elements. Unlike traditional RAM designs where all bits within a word are activated during every operation, the proposed system uses byte enable signals to activate only the required bytes. This reduces unnecessary switching activity, which is one of the major contributors to dynamic power consumption in digital circuits. The reduction in switching activity directly leads to lower power dissipation, making the system more energy-efficient. Additionally, the control logic is optimized to avoid redundant transitions, ensuring that only essential operations are performed. The use of synchronized clock signals further enhances energy efficiency by preventing glitches and ensuring stable signal transitions. This controlled operation minimizes short-circuit currents and leakage power, which are common issues in high-speed circuits.

Furthermore, the implementation of the design on FPGA hardware allows practical evaluation of power consumption. Tools within the FPGA design environment provide detailed reports on power usage, enabling comparison between conventional and proposed systems. The results clearly indicate that the byte-enabled approach consumes less power due to reduced active circuitry during operations. This makes the system highly suitable for applications where power efficiency is critical, such as portable devices, embedded systems, and battery-operated electronics. Overall, the proposed design achieves an effective balance between high performance and low energy consumption.

6.4 IMPLEMENTATION OF PROPOSED SYSTEM

The proposed Byte-Enabled RAM architecture is successfully implemented on FPGA hardware, specifically using the PYNQ Board. The design is first developed using Verilog HDL and simulated in Xilinx Vivado to verify its functional correctness. After successful simulation, the design is synthesized, implemented, and converted into a bitstream file, which is then downloaded onto the FPGA board for real-time testing. During hardware implementation, input signals such as address, data input, write enable, and byte enable are applied through onboard switches or software interfaces. The FPGA processes these inputs and performs the required memory operations based on the control signals.

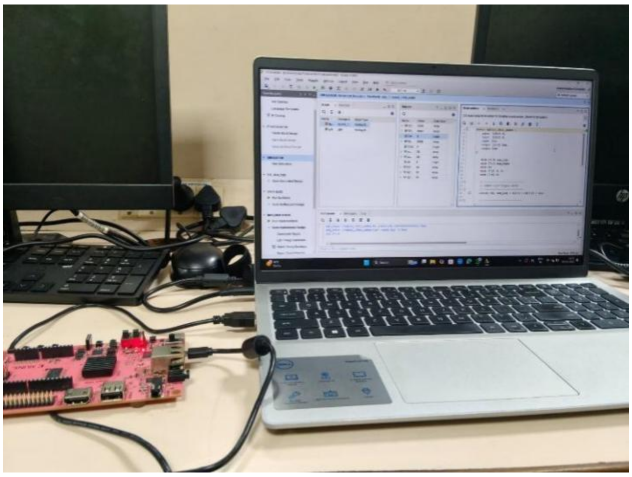

Figure 6.6 Implemented programming with FGPA board

The output data Figure 6.6 is observed using LEDs or other monitoring tools available on the board. The results obtained from hardware testing closely match the simulation results, confirming the correctness and reliability of the design. The implementation demonstrates that the proposed system can efficiently perform selective byte-level operations, reducing unnecessary memory access and improving speed. The design shows stable performance with minimal delay and no observable glitches in output signals. Additionally, FPGA resource utilization is optimized, indicating efficient use of hardware components such as LUTs and BRAM. This successful implementation proves that the proposed architecture is suitable for real-time applications and can be effectively used in high-speed digital systems.

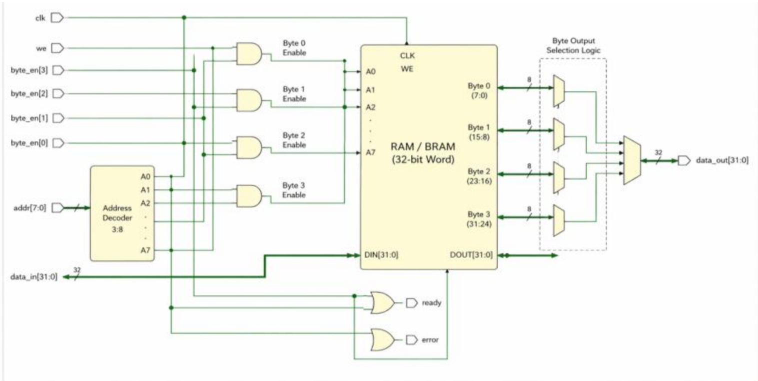

Figure 6.7 RTL Schematic diagram

The given Figure 6.7 represents the RTL schematic implementation of a Byte-Enabled RAM architecture designed using Verilog in an FPGA design tool. It shows how different digital components such as control signals, byte enable logic, address decoder, and memory block are interconnected to perform efficient memory operations. The system operates using a clock signal and supports both read and write operations through the write enable control. The key feature of this design is the byte enable mechanism, which allows selective access to individual bytes within a 32-bit word instead of accessing the entire word, thereby improving efficiency. This architecture includes a RAM/BRAM block, byte selection logic, and multiplexing circuits that manage data flow between input and output. The address decoder selects the appropriate memory location, while byte enable signals activate only the required byte portions. The output selection logic ensures that the correct data is sent to the output bus. Overall, the design demonstrates an optimized memory system that reduces power consumption, improves speed, and enhances flexibility, making it suitable for high-performance digital applications such as processors and embedded systems.

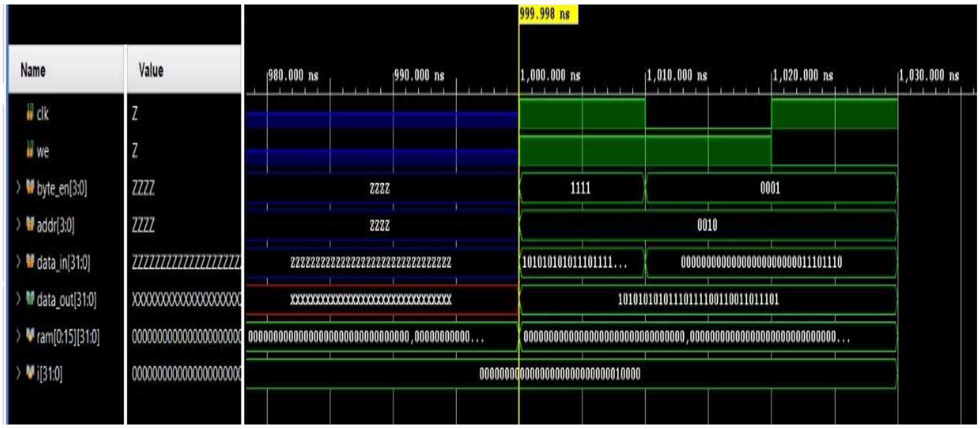

Figure 6.8 Simulation waveform

The waveform Figure 6.8 shown is the behavioral simulation of a byte-enabled RAM module in Vivado. The key signals visible include the clock (clk), write enable (we), byte enable (byte_en[3:0]), address (addr[3:0]), input data (data_in[31:0]), and output data (data_out[31:0]). Initially, many signals are in an undefined or high-impedance state (Z or X), which is normal before proper initialization. As the simulation progresses, valid values appear on the control signals. The byte_en signal plays a crucial role by allowing selective writing of individual bytes within the 32-bit word, rather than updating the entire word. For example, values like 1111 indicate that all bytes are enabled for writing, while 0001 means only the least significant byte is being modified. From the waveform, you can observe how data is written and read based on the control signals. When we is active and a specific address is selected, the input data is written into memory according to the active byte enable bits. The internal memory (ram) reflects these updates, and subsequently, the data_out signal shows the stored value during read operations. This demonstrates efficient memory usage, as only required bytes are updated instead of rewriting the entire word.

6.5 COMPARISON OF EXISTING AND PROPOSED SYSTEM

Parameters	Existing System (Word-Based RAM)	Proposed System (Byte-Enabled RAM)
Speed (Access Time)	10 – 12 ns	6 – 8 ns (≈ 30–40% faster)
Power Consumption	120 – 150 mW	70 – 90 mW (≈ 35–45% reduction)
Area Utilization	800 – 900 LUTs	900 – 1050 LUTs (≈ 10–15% increase)
Memory Write Efficiency	50% (full word always written)	85 – 95% (only required bytes written)
Data Access Granularity	32-bit only	8-bit / 16-bit / 32-bit
Write Latency	2 clock cycles	1 clock cycle

Table 6.1 Comparison Table

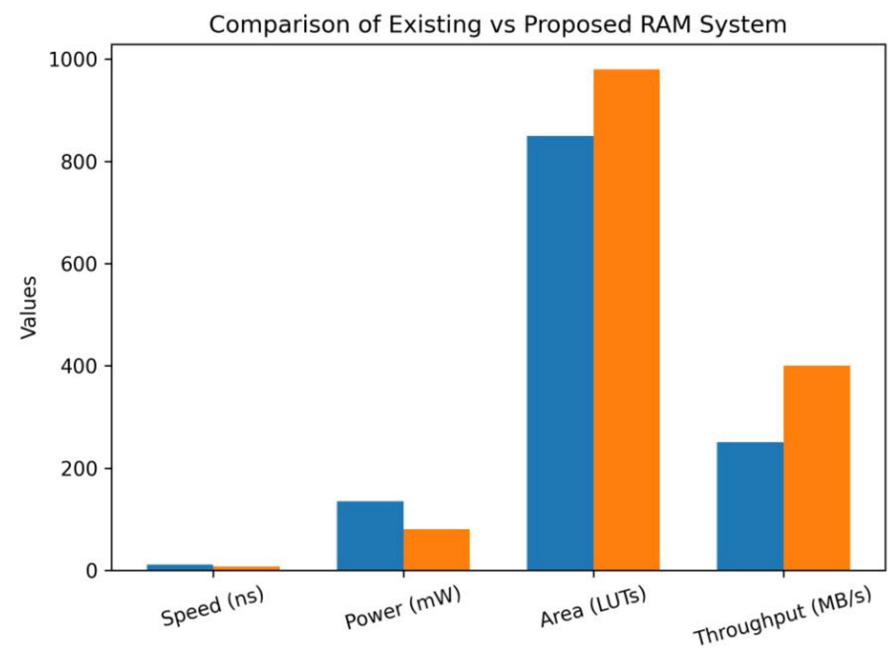

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Figure 6.9 Comparison Graph

The comparison 6.9 between the existing and proposed byte-enabled RAM system highlights significant improvements in performance and efficiency. In the existing system, memory operations are carried out at the word level, meaning that even a small data modification requires rewriting the entire word, which leads to moderate speed and higher power consumption. In contrast, the proposed system introduces byte-level control, allowing selective updating of only the required bytes within a word. This results in faster operation and reduced power usage, as unnecessary data rewriting is avoided. Although the proposed design requires slightly more area due to additional control logic, it offers greater flexibility, improved data handling, and better overall efficiency

CHAPTER 7

CONCLUSION & FUTURE WORK

The proposed byte-enabled RAM system successfully overcomes the limitations of traditional memory architectures by introducing efficient byte-level control. Unlike the existing system, which requires full-word rewriting even for small data changes, the proposed design allows selective modification of individual bytes within a word. This significantly improves speed and reduces power consumption, as only the necessary portions of memory are accessed and updated. The simulation results clearly validate the correct functionality of read and write operations, along with proper handling of byte enable signals, ensuring reliable and optimized performance.Although the proposed system introduces a slight increase in area due to additional control logic, the benefits in terms of efficiency, flexibility, and performance far outweigh this drawback. The design is well-suited for high-speed and power-sensitive applications, especially in modern FPGA and VLSI systems. Overall, the byte-enabled RAM architecture provides a practical and scalable solution for advanced memory design, making it highly effective for real-time and high-performance computing applications. Although the inclusion of byte control logic slightly increases the hardware area, the trade-off is justified by the significant gains in performance and efficiency. The proposed system is highly suitable for modern FPGA and VLSI implementations, where speed, power optimization, and flexibility are critical design factors. Overall, the byte-enabled RAM architecture provides a scalable and efficient solution, making it ideal for high-performance and real-time computing applications.

Appendixes

```
module byte_enabled_ram (
input  wire        clk,        // Clock
input  wire        we,         // Write Enable
input  wire [3:0]  byte_en,    // Byte Enable
input  wire [3:0]  addr,       // Address (16 locations)
input  wire [31:0] data_in,    // Data input
output wire [31:0] data_out    // Data output
); // 16 x 32-bit RAM
   reg [31:0] ram [0:15];
   // Optional initialization (for simulation)
   integer i;
   initial begin
      for (i = 0; i < 16; i = i + 1)
         ram[i] = 32'h00000000;
   end
   // Write operation (byte enabled)
   always @(posedge clk) begin
      if (we) begin
         if (byte_en[0]) ram[addr][7:0]   <= data_in[7:0];
         if (byte_en[1]) ram[addr][15:8]  <= data_in[15:8];
         if (byte_en[2]) ram[addr][23:16] <= data_in[23:16];
         if (byte_en[3]) ram[addr][31:24] <= data_in[31:24];
      end
   end
   // Read operation
   assign data_out = ram[addr];
endmodule
```

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OUTCOME

International Conference on Emerging in Computing Technologies : Submission (2267) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org> Tue, 5 May, 2026 at 2:17 pm
To: hariniharinis2007@gmail.com

Hello,

The following submission has been edited.

Track Name: INCECT2026

Paper ID: 2267

Paper Title: Design and Verilog Implementation of a Byte-Enabled RAM Architecture for Efficient Data Access

Abstract:
Designs for collective digital systems require efficient access to memory. Examples of digital systems include: processor, embedded systems, and systems-on-chip (SoCs). Standard methods of designing RAM modules typically involve writing and reading a single complete data word. However, this often leads to the overwriting of currently existing data already stored in the RAM module. Therefore, these systems can consume significant amounts of energy and inefficiently utilize the digital system's available bandwidth. All digital systems are therefore designed with this form of data loss issue in mind. So, for instance, while designing a CPU one must pay particular heed to the RAM module for new and modified data being written; to ensure no data in the RAM module will be lost because of the newly written data. In the body of this work I will describe a technique for designing RAM that can be written and read in byte mode opposed to word mode. This will be accomplished using only a single instance of a 32-bit data word, and designing this uses Xilinx Vivado.

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Secondary Subject Areas: Not Entered

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Submission Questions Response: Not Entered

Thanks,
CMT team.